

UBT624: MEDICAL BIOTECHNOLOGY

L	T	P	Cr
3	0	0	3.0

Course Objective: To enlighten the knowledge of the students on different areas of Medical Biotechnology. To train the students in a hospital-based setup and familiarize them with the clinical diagnostics of diseases.

Syllabus

Introduction: History and scope of medical biotechnology, current status and future prospects.

Classification of genetic diseases: Chromosomal disorders – Numerical disorders e.g. trisomies & monosomies, Structural disorders e.g deletions, duplications, translocations & inversions, Chromosomal instability syndromes. Gene controlled diseases – Autosomal and X-linked disorders, Mitochondrial disorders.

Molecular basis of human diseases: Pathogenic mutations Gain of function mutations: Oncogenes, Huntingtons Disease, Pittsburg variant of alpha 1 antitrypsin. Loss of function - Tumour Suppressor. Genomic. Dynamic Mutations - Fragile- X syndrome, Myotonic dystrophy. Mitochondrial diseases.

Gene therapy: *Ex-vivo*, *In vivo*, *In situ* gene therapy, Strategies of gene therapy: gene augmentation Vectors used in gene therapy Biological vectors – retrovirus, adenoviruses, Herpes Synthetic vectors– liposomes, receptor mediated gene transfer. Gene therapy trials – Familial Hypercholesterolemia, ADA, AIDS, Cystic Fibrosis, Solid tumors.

Nucleic acid-based Therapy: Gene silencing technology, siRNA, Aptamers, antisense oligodeoxynucleotides (AS-ODN), Ribozymes, Peptide Nucleic Acids.

Recombinant & Immunotherapy: Clinical applications of recombinant technology; Erythropoietin; Insulin analogs and its role in diabetes; Recombinant human growth hormone; Streptokinase and urokinase in thrombosis; Recombinant coagulation factors, Monoclonal antibodies and their role in cancer; Role of recombinant interferons; Immunostimulants; Immunosuppressors in organ transplants; Role of cytokine therapy in cancers.

Clinical management and Metabolic syndrome: PKU, Familial Hypercholesterolemia, Rickets, ADA, Congenital hypothyroidism.

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

Course Learning Objectives (CLO)

The students will be able to:

1. Explain insights about genetic diseases and also about the molecular aspects related to human disease.
2. Gain new insights into molecular mechanisms of nucleic acid and gene therapy.
3. Gain knowledge about therapeutic recombinant proteins and immunotherapy for the treatment of different diseases.

Text Books

1. Diagnostic and Therapeutic Antibodies (Methods in Molecular Medicine by Andrew J.T. George (Editor), Catherine E. Urch (Editor) Publisher: Humana Press; edition (2000)
2. Molecular Diagnosis of Infectious Diseases (Methods in Molecular Medicine) by Jochen Decker, U. Reischl Amazon

Reference Books

1. Human Molecular Genetics by T. Strachan, Andrew Read Amazon Sales Rank.

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes)	30

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